

SURV 626: Sampling

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Unit 1: Preliminaries

Course Description

Content

- Design vs. Analysis
 - Techniques (SRS, stratification, clustering, etc.)
 - Sampling variance estimation
- Finite population
- Design-based vs. Model-based inference
- Descriptive vs. Analytic
- Statistics
 - Means, proportions, totals
 - Variances
 - Correlations, regression coefficients

Unit 1: Preliminaries

Course Perspectives

- A. Design
- B. Typology & vocabulary
- C. Notation
- D. Sampling distribution

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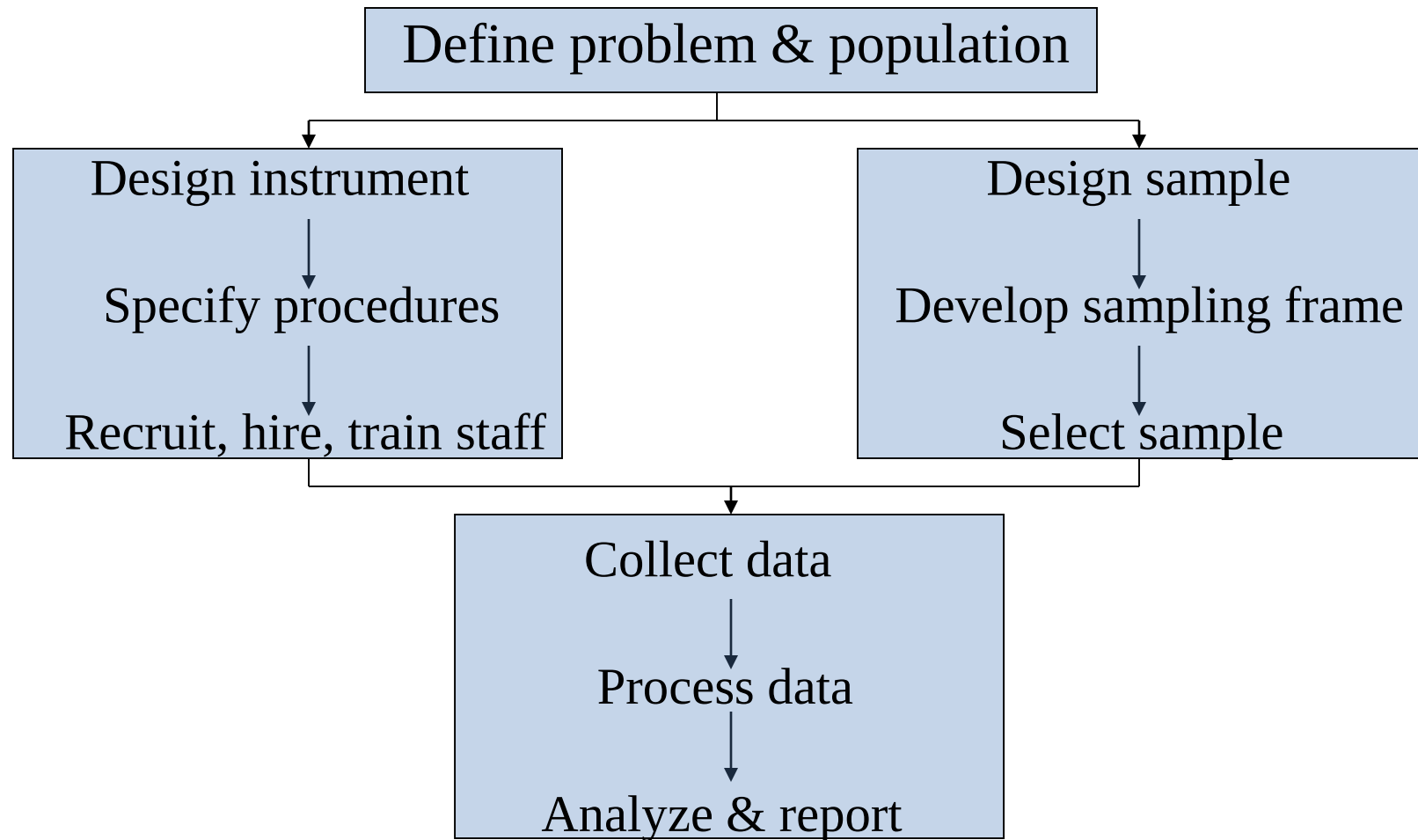
A. Design

- Research design
 - Experiments
 - Dependent variables
 - Factors
 - Control or randomization of disturbing variables
 - Quasi-experimental: observational
 - Survey samples: observational
- Design characteristics
 - Realism
 - Randomization
 - “Representation”

Sampling designs

- Probability sampling designs
 - Simple random sampling
 - Stratified sampling
 - Cluster sampling
 - Stratified cluster sampling, stratified multistage sampling
- Non-probability sampling designs
 - Convenience, haphazard
 - Snowball, Respondent Driven Sampling (RDS)
 - Volunteer
 - Quota sampling

Sample Surveys



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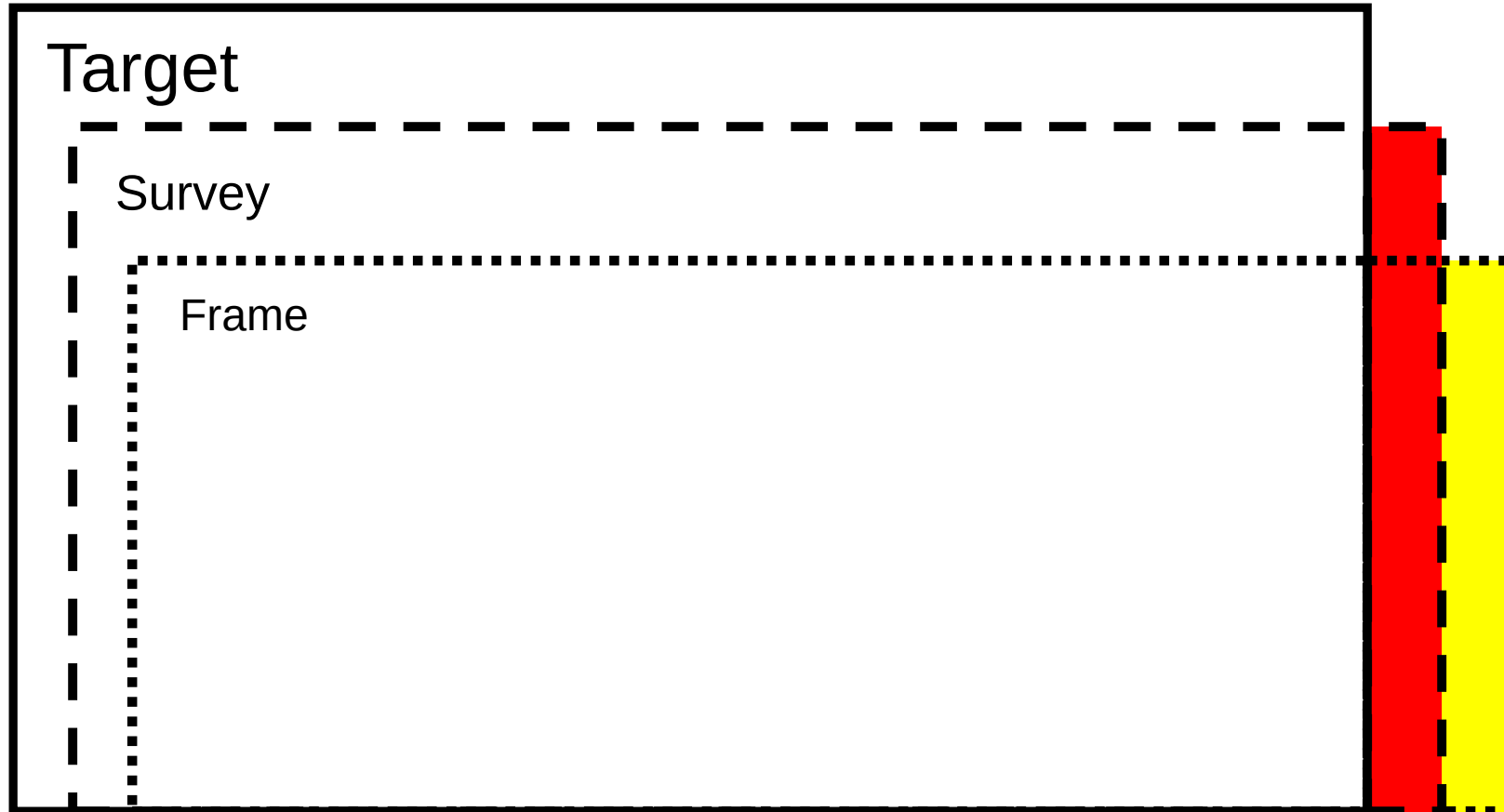
Course Perspectives

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B. Typology & vocabulary

- Finite population
- Large population & sample
- Randomization
- “Representative”
- Efficient (precision per unit cost)

Population



Survey Sampling Design Typology

Control	Unit	Chance	
		<i>Equal</i>	<i>Unequal</i>
<i>None</i>	<i>Element</i>	SRS or systematic	Non-epsem element
	<i>Cluster</i>	Cluster	Non-epsem cluster
<i>Grouped</i>	<i>Element</i>	Proportionate stratified random	Disproportionate stratified random
	<i>Cluster</i>	Proportionate stratified cluster	Disproportionate stratified cluster

Additional vocabulary

- Observation
 - Coverage
 - Response
- Quality
 - Reliability, variance
 - Validity, bias
- Properties
 - Unbiasedness
 - Precision
 - Accuracy (mean square error)
- Sampling distribution
 - Expectation
 - Standard error
- Inference
 - Central limit theorem
 - Confidence interval
 - Normal & t distributions
- Non-probability sampling
 - Volunteer
 - Purposive, judgment
 - Quota
- Measurability
- Sample design
 - Plan
 - Estimation

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C. Notation: Population

- N elements labels $i = 1, 2, \dots, N$
- Characteristics denoted $Y_1, Y_2, \dots, Y_N$

$$Y = Y_1 + Y_2 + \dots + Y_N$$

$$\bar{Y} = Y/N$$

$$P = Y/N$$

$$S^2 = \frac{1}{N-1} \sum_{i=1}^N (Y_i - \bar{Y})^2$$

$$S^2 = \left(\frac{1}{N-1} \right) \left(\sum_{i=1}^N Y_i^2 - Y^2/N \right)$$

$$S^2 = \left(\frac{N}{N-1} \right) P(1-P)$$

Sample

- Sample n elements from N , $i = 1, 2, \dots, n$
- Values denoted as $y_1, y_2, \dots, y_n$

$$y = y_1 + y_2 + \dots + y_n$$

$$\bar{y} = y/n$$

$$p = y/n$$

$$s^2 = \frac{1}{n-1} \sum_{i=1}^n (y_i - \bar{y})^2$$

$$s^2 = \left(\frac{1}{n-1} \right) \left(\sum_{i=1}^n y_i^2 - y^2/n \right)$$

$$s^2 = \left(\frac{n}{n-1} \right) p(1-p) \approx p(1-p)$$

$$f = n/N \text{ and } F = f^{-1} = N/n$$

Unit 1: Preliminaries

Course Perspectives

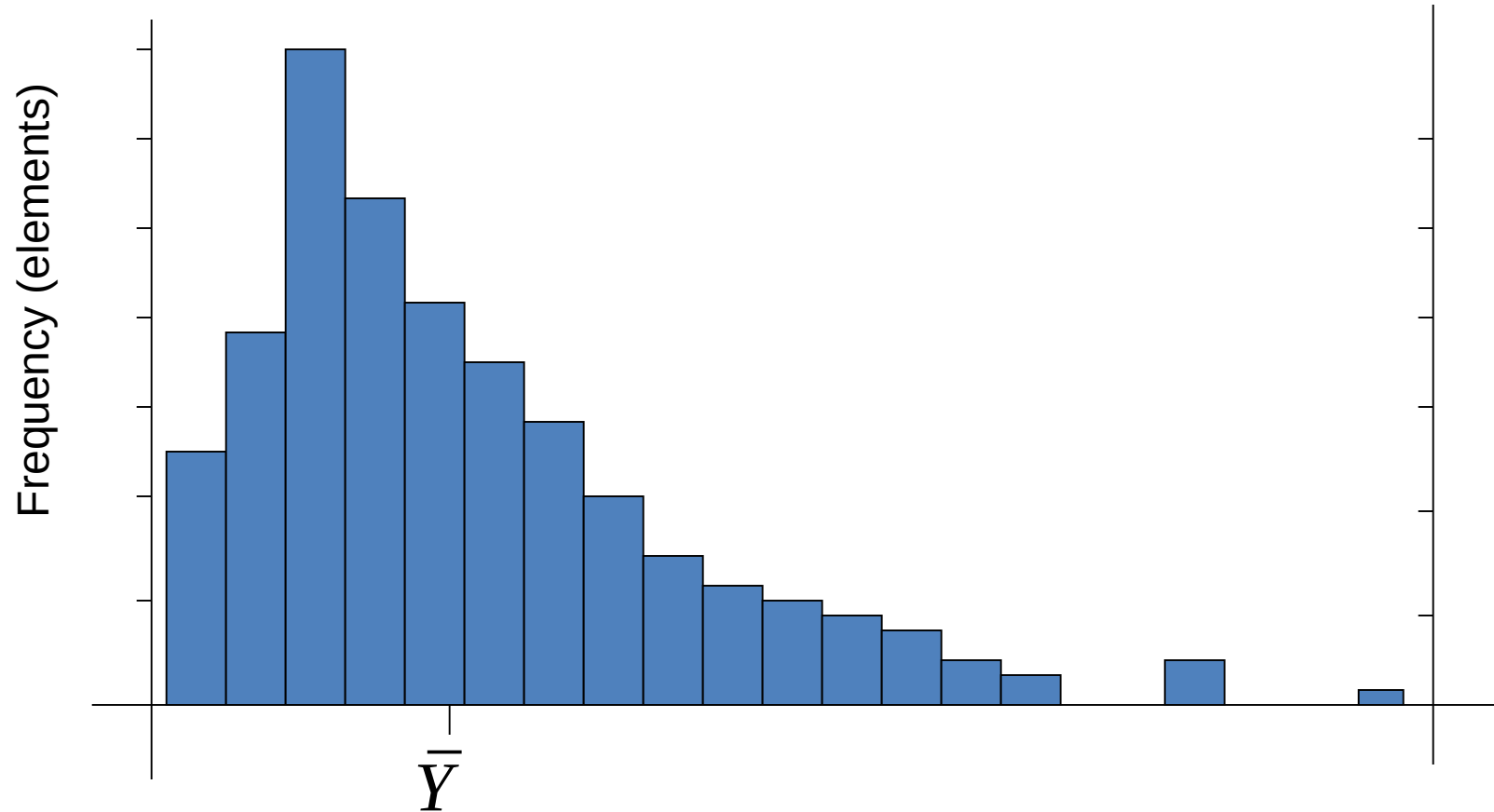
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D. Population distribution



Sampling distribution

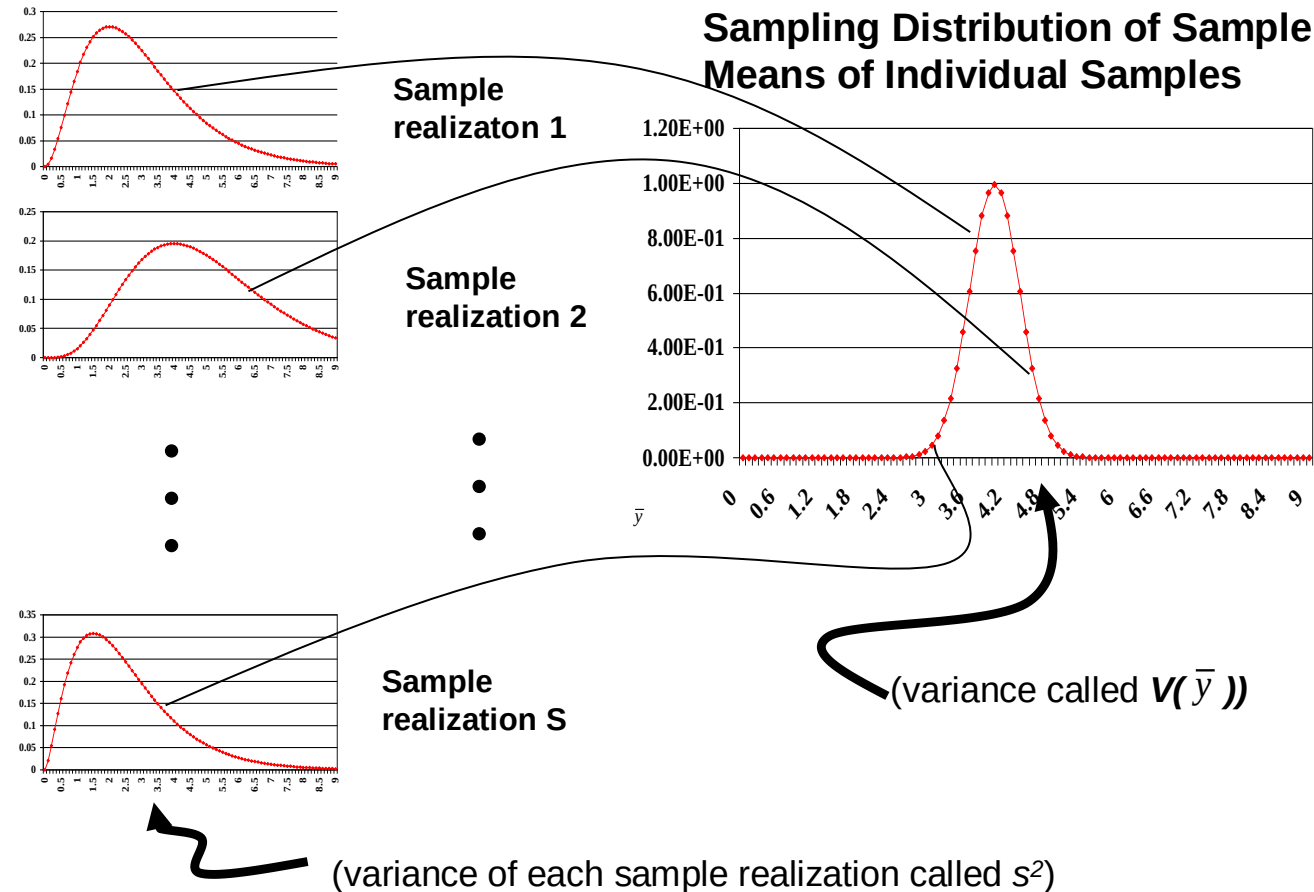


Figure 4.2 Distributions of y Variable from Sample Realizations Samples and the Sampling Distribution of the Mean